



Darukaa.Earth: Full-Stack Developer Hackathon Challenge

Welcome to the Darukaa.Earth Full-Stack Developer Hackathon! This challenge is designed to evaluate your technical skills, product mindset, and ability to communicate your work effectively. We are looking for talented engineers who can not only write clean code but also design elegant systems and build products with a clear purpose.

Challenge Overview: Business Requirements

Your task is to build a full-stack, geospatial data analytics platform. The application, tentatively named "Darukaa.Earth," should serve as a dashboard for managing and visualizing carbon and biodiversity projects.

Core User Stories:

- **As an administrator**, I want to create a new project and add multiple geographical sites to it.
- **As an administrator**, I want to view all projects and sites on an interactive map.
- **As an administrator**, I want to click on a specific site to view detailed analytics and performance over time.

Key Features:

- **User Authentication:** A basic user registration and login system.
- **Project Management:** A dashboard to create and view projects.
- **Geospatial Data:** The ability to add new sites by drawing polygons on a map.
- **Data Visualization:** The application must display analysis results in a clear, interactive format.
- **Automated code quality check:** Use tools/libraries to support automated code quality check to maintain the stability of the platform.

Technical Requirements: Tools & Tech Stack

We expect you to use a modern and industry-standard technology stack. The overall project architecture should adhere to the following tech stack.

- **Frontend:**
 - **Framework:** React
 - **Mapping:** Mapbox GL JS for interactive maps.
 - **Charting:** Highcharts or Chart.js for data visualization.
- **Backend:**
 - **Language & Framework:** Python with Flask/Django/ Fast API.
 - **Database:** PostgreSQL with the PostGIS extension for handling geospatial data.

- **Authentication:** JWT-based user authentication.
- **CICD Pipeline & Developer Experience:**
 - **Version Control:** Git on a private GitHub repository.
 - **CI/CD:** Use **GitHub Actions** to automate your pipeline.
 - **Deployment:** The application must be automatically deployed to a publicly accessible URL on platforms like Heroku, Vercel, or Render.com.
 - **Code Quality:** Implement **pre-commit hooks** using tools like **Husky** and **lint-staged** to automatically format your code (e.g., with Prettier) and run linting checks before every commit. This is a crucial requirement.

Note: There are no limitations on datasets and mocks you would want to use in the project, feel free to use any datasets and document why this choice was made.

Submission Deliverables

To be considered for the position, you must submit the following:

1. **GitHub Repository Link:** A private GitHub repository containing your full-stack application code. The repository must be clean and have a logical commit history. Access should be granted to the hiring team.
2. **Live Demo URL:** A public URL where we can access and test your working application.
3. **README.md File:** A comprehensive document in your repository that includes:
 - A clear explanation of the high-level architecture.
 - A breakdown of your database schema.
 - Instructions on how to set up the environment and run the application locally.
 - A detailed explanation of your CI/CD pipeline, including your GitHub Actions workflow configuration.

Evaluation Criteria

Your submission will be evaluated on a holistic basis, not just on your code. We will score your submission across three key areas:

1. Technical Excellence

- **Code Quality:** Is the code clean, readable, well-organized, and maintainable?
- **System Architecture:** Did you design a logical and scalable full-stack system with appropriate database schema?

- **CI/CD & Developer Experience:** Is the CI/CD pipeline robust and correctly configured? Are the commit hooks effectively enforcing code standards?
- **Technical Execution:** Did you correctly integrate and utilize the specified technologies like Mapbox and Highcharts?

2. Product Mindedness

- **Problem-Solving:** Did you understand the core business requirements and deliver a solution that addresses them effectively?
- **UI/UX Design:** Is the application intuitive and pleasant to use? Is the data visualization clear and easy to understand?
- **Trade-offs:** Did you make sensible technical choices and can you clearly articulate the reasoning and trade-offs behind them?

Documents Submission Guidelines

Submit one Word document (.docx) through the job where you applied, using the **My Jobs** page / **Applied Job** page and its document-submission option.

The Word document must include:

- 1 GitHub repository link for the completed project.
- 2 Live demo URL, where applicable.
- 3 A brief README.md overview covering the architecture, database/schema, local setup, and CI/CD details.
- 4 Any other links, credentials, or notes required to review and run the submission.

Repository access instructions

If the GitHub repository is private, grant access to the following accounts and include the repository link in the Word document:

- ankita.dasgupta@darukaa.com
- harsh.kumar@darukaa.com
- utkarsh.gauniyal@darukaa.com
- guneet.mutreja@darukaa.com

If the GitHub repository is public, provide the repository link only; access invitations are not required.

All submission details and links must be included in the Word document submitted through the applied-job page.